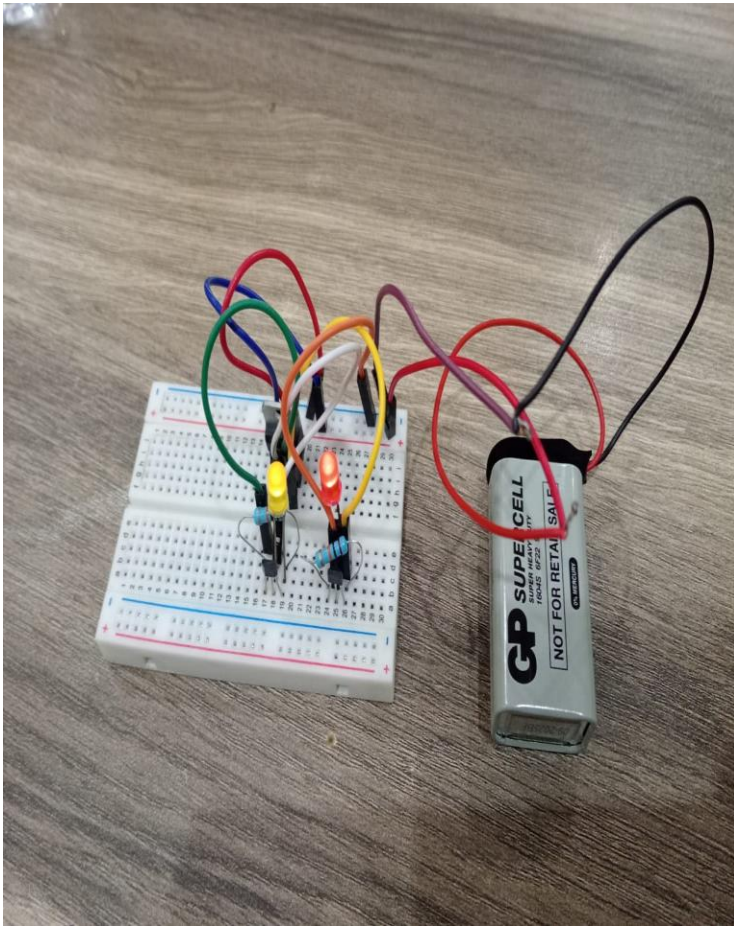




MAGNETIC POLARITY DETECTOR USING HALL-EFFECT SENSORS

SUPERVISED BY:

DR. Nour Eldin

Eng. Ehab Ahmed

TEAM MEMBERS:

-Mostafa Hatem

-Yousef Abdallah

-Mriam Mansour

-Menna Alrahman

Mohamed

1. Project Summary

This project is a compact Magnetic Polarity Detector designed to identify the North and South poles of any permanent magnet. Using two A3144 Hall-effect sensors mounted in opposite orientations, the circuit converts magnetic polarity into visual LED indicators.

2. Scientific Background

2.1 Hall Effect Theory

When charge carriers move through a conductor and a magnetic field is applied perpendicular to their motion, they experience the Lorentz force:

$$\mathbf{F} = q (\mathbf{v} \times \mathbf{B})$$

This pushes electrons to one side, creating a Hall Voltage.

Hall Voltage Formula: $V_H = IB / (nqt)$

3. Hall Sensor Specifications (A3144)

- Operating Voltage: 4.5–24 V
- Output Type: Digital (open collector)
- Activation: 30–80 Gauss

4. Principle of Polarity Detection

Two A3144 sensors oriented opposite each other detect different poles.

5. Circuit Design

5.1 Voltage Regulation

Using 7805 regulator: $P = (9V - 5V) \times 0.04A = 0.16 \text{ W}$

5.2 Output Response

LED current (no magnetic field): $I = (5V - 2V) / 10k\Omega = 0.3 \text{ mA}$

6. Circuit Operation Summary

North pole → Sensor-A output LOW → LED OFF

South pole → Sensor-B output LOW → LED OFF

7. Testing Procedure

1. Power on – LEDs ON
2. Bring South pole → Red LED OFF
3. Bring North pole → Green LED OFF

8. Applications

- Magnet sorting systems
- Motor polarity detection
- Robotics
- Education tools

9. Extensions

- Digital polarity meter (Arduino + OLED)
- Shaft rotation detection (3 Hall sensors)
- Magnetic field mapping tool

Conclusion

This system reliably identifies magnetic poles and can be expanded for advanced sensing applications.